Loss · normal → 90m / 6h / 10h

One number per target time, then the mean over $t \in \{1, 2, 3\}$. The cell question has weight 0.

$$\mathcal{L} = \text{mean}_t(\mathcal{L}_{\text{var}} + 0.25\,\mathcal{L}_{\text{abs}} + 0.1\,\mathcal{L}_{\text{NCE}} + 0.1\,\mathcal{L}_{\text{mask}})$$

$\mathcal{L}_{\text{var}}$ — $\lambda = 1$

$$\text{mean}_{g \in P}\ \tfrac{1}{768}\,\|\hat{z}_g - z^\star_g\|^2$$

P: gene is on at time t

and the slot is outside the 19 masks

$z^\star_g$ is its row of H_{tgt}

both vectors raw, in $\mathbb{R}^{768}$

$\mathcal{L}_{\text{abs}}$ — $\lambda = 0.25$

$$\text{mean}_{g \in A}\ \tfrac{1}{768}\,\|\hat{z}_g - a_{\text{null}}\|^2$$

A: gene is off at time t

and the slot is outside the 19 masks

$a_{\text{null}} \in \mathbb{R}^{768}$ is fixed

$\|a_{\text{null}}\| = \sqrt{768}$

$\mathcal{L}_{\text{NCE}}$ — $\lambda = 0.1$

$$\text{CE}(\text{softmax}(S/0.1))$$

$S_{ij} = \hat{u}_i \cdot u^\star_j$

$\hat{u}$, $u^\star$ are L2-normalized

same slots as $\mathcal{L}_{\text{var}}$, over the batch

positive: that query's own target row

$\mathcal{L}_{\text{mask}}$ — $\lambda = 0.1$

$$\text{CE}(W\hat{z}/\|\hat{z}\|,\ g)$$

the 19 mask slots

$W : \mathbb{R}^{768} \to \mathbb{R}^{V}$

V is the local gene vocabulary

label is the hidden gene id

Where the 128 gene questions go

On at time t → $\mathcal{L}_{\text{var}}$ and $\mathcal{L}_{\text{NCE}}$.

Off at time t → $\mathcal{L}_{\text{abs}}$.

The 19 mask slots → $\mathcal{L}_{\text{mask}}$ alone.

On and off are read from the target, after the quiz is drawn.